TOPIC 1 FUNCTIONS

1.1 Functions

1.1.1 Definition

A function f from set X to set Y is a relation that assigns to each element (the independent variable) of the set X to a unique element (the dependent variable) of the set Y using a rule.

It is represented symbolically as

① $y = f(x)$ y is a function of x

② $f: X \rightarrow Y$ function f maps X onto Y

1.1.2 Domain and range

Domain is defined as all the possible REAL values of the independent variable that makes the function defined.

Range is defined as all the possible REAL values of the dependent variable.

Complex numbers

$z = a + bi$

$\text{Re}(z)$ $\quad$ $\text{Im}(z)$

Example 1

Determine the domain of the following functions of $y = f(x)$.

(a) $f(x) = x + 3$ (polynomials)

S: D_f: $x \in R$

I: D_f: $(-\infty, +\infty)$

(b) $f(x) = \frac{1}{x+3}$ (rational function)

$x + 3 \neq 0$

S: D_f: $x \neq -3$.

I: D_f: $(-\infty, -3), (-3, +\infty)$

(c) $f(x) = \sqrt{x+3}$ (root function)

$x + 3 \geq 0$

S: D_f: $x \geq -3$

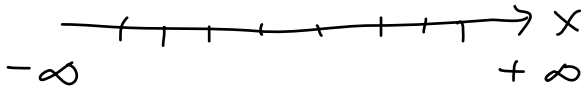
I: D_f: $[-3, +\infty)$.

Set notation	Interval notation
$x \in R$	$(-\infty, +\infty)$
$x < a$	$(-\infty, a)$
$x \leq a$	$(-\infty, a]$
$x > a$	$(a, +\infty)$
$x \geq a$	$[a, +\infty)$

(d) $f(x) = 4x^2 + 7x - 19$ (polynomial : $y = a_n x^n + a_{n-1} x^{n-1} + \dots$)

S: $D_f : x \in \mathbb{R}$

I: $D_f : (-\infty, +\infty)$



(e) $f(x) = \frac{x+1}{x^2+4x+3}$ numerator (rational) denominator $\neq 0$

$x^2 + 4x + 3 \neq 0$

$(x+1)(x+3) \neq 0$

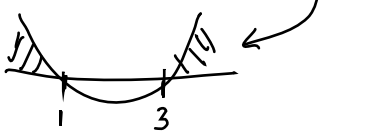
S: $D_f : x \neq -3, x \neq -1$.

I: $D_f : (-\infty, -3), (-3, -1), (-1, +\infty)$.

(f) $f(x) = \sqrt{x^2 - 4x + 3}$ (square root)

$x^2 - 4x + 3 \geq 0$

$(x-1)(x-3) \geq 0$



S: $D_f : x \leq 1, x \geq 3$

I: $D_f : (-\infty, 1], [3, +\infty)$

1.2 Exponential & Logarithmic Functions

1.2.1 Definition of exponential function

exponent – power (index)

If b and c are positive numbers where $b > 0$, $b \neq 1$, then

$$f(x) = b^x$$

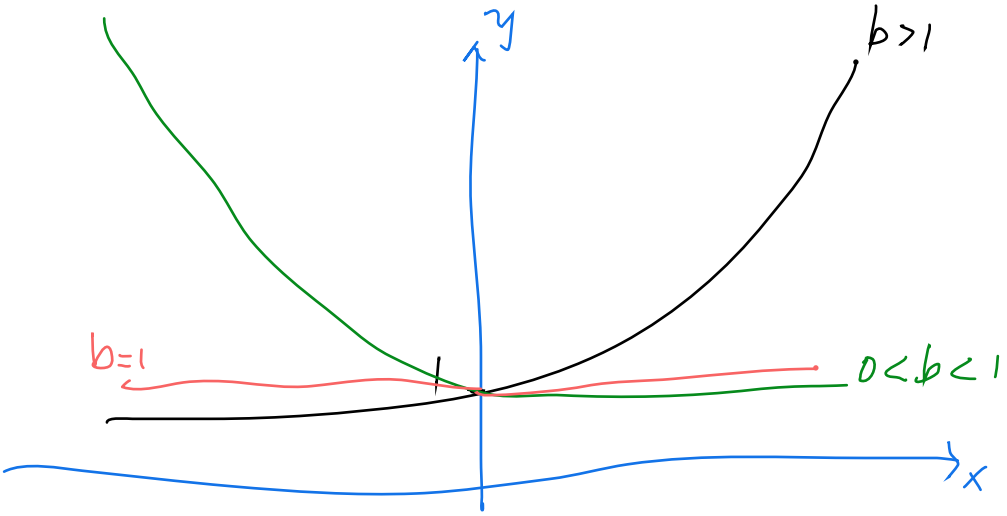
is exponential function.

Natural exponential : $f(x) = e^x = \exp(x)$

Calculator

$e \approx 2.71828....$

1.2.2 Graph of exponential function



1.2.3 Definition of logarithmic function

If b and c are positive numbers where $b > 0$, $b \neq 1$ and $b^x = c$, then the number x is the logarithm of c to the base b and is written as

$$x = \log_b c$$

base

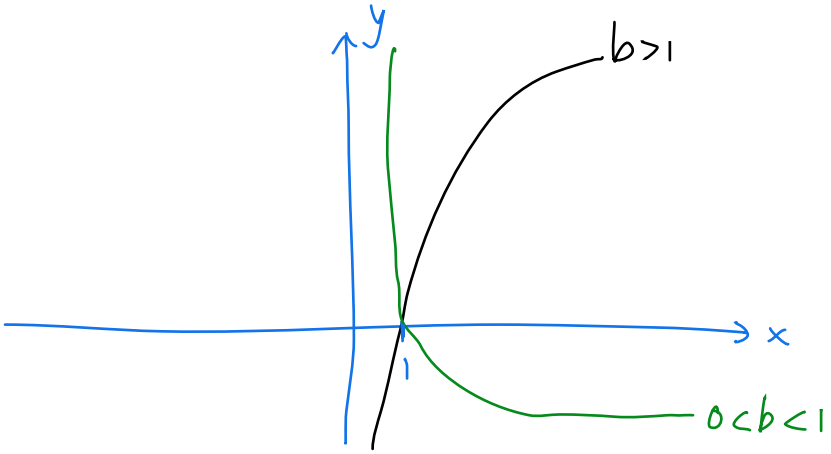
$\log_{10} x = \log x = \lg x$

Natural log: $\log_e x = \ln x$

($\ln x$)

($\ln v$)

1.2.4 Graph of logarithmic function



Example 2

(a) Write in the form e^m where $m > 0$: $e^n \times (e^n)^2 \div e^{-3n} = e^{n+2n-(-3n)} = e^{6n}$ ✱

(b) Write $\ln x + \ln(x^2y) - \ln(y^3)$ as a single term.

$= \ln \dfrac{x(x^2y)}{y^3}$

$= \ln \left(\dfrac{x^3}{y^2}\right)$ ✱

Example 3

Solve for x:

(a) $\log_3 27 = x$

$3^x = 27$

$= 3^3$

$\therefore x = 3$ ✱

(b) $\log_x 64 = 3$

$x^3 = 64$

$= 4^3$

$\therefore x = 4$ ✱

(c) $\log_2 64 = x$

$2^x = 64$

$= 2^6$

$\therefore x = 6$ ✱

(d) $2^x = 3$

$\lg 2^x = \lg 3$

$x \lg 2 = \lg 3$

$x = \frac{\lg 3}{\lg 2} = 1.585$ ✱

(e) $3^{x+1} = 5$

$(x+1)\lg 3 = \lg 5$

$x + 1 = \frac{\lg 5}{\lg 3}$

$\therefore x = 0,465$ ✱

(f) $\log 50 + \log x = 2 + \log(x-1)$ (Ans : $x = 2$)

$\log 50 + \log x - \log(x-1) = 2$

$\log \frac{50x}{x-1} = 2$

$50x = 10^2(x-1)$

$50x = 100$

$x = 2$ ✱

(g) $2\log 3 + \log 2x = \log(3x + 1)$ (ans : $x = \frac{1}{15}$)

$\log 3^2(2x) = \log(3x+1)$

$18x = 3x + 1$

$x = \frac{1}{15}$ ✱

Rules of log

$\log a + \log b = \log(ab)$

$\log a - \log b = \log\left(\frac{a}{b}\right)$

$\log a^b = b \log a$

(h) $3^{x+1} = 2^{x+2}$ (ans : 0.7093)

$(x+1)\lg 3 = (x+2)\lg 2$

$x\lg 3 + \lg 3 = x\lg 2 + 2\lg 2$

$x(\lg 3 - \lg 2) = 2\lg 2 - \lg 3$

$x = \frac{0.1249}{0.1761}$

$= 0.7093$ ✱

(i) $2^{x-3} = 5^{2x-4}$ (ans : 1.726)

$(x-3)\lg 2 = (2x-4)\lg 5$

$x\lg 2 - 2x\lg 5 = 3\lg 2 - 4\lg 5$

$x = \frac{0.9031 - 2.7959}{0.3010 - 1.3979}$

$= 1.7256$ ✱